

Deltalogix Insights

THE STATE OF DISTRIBUTED LEDGER TECHNOLOGIES

Blockchain and its peers

2023

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INTRODUCTION

Distributed Ledger Technologies (DLTs) have revolutionized the landscape of digital transactions by enabling decentralization, immutability, and high-security standards. DLTs offer a new paradigm for recording, validating, and managing commercial transactions, leading to innovative trust, security, and transparency frameworks.

DLTs can record, store, and exchange digital information among consenting parties without a central trust authority. They allow secure and accurate data storage using cryptography, resulting in an immutable database governed by integrated network rules. There exist various types of DLTs, including Blockchain, Hashgraph, Directed Acyclic Graph (DAG), Holochain, and TEMPO, each with its unique features. To maintain data integrity and security, these technologies employ various consensus mechanisms, with Proof of Work (PoW) and Proof of Stake (PoS) being the most prevalent. Moreover, innovative variants such as Delegated Proof of Stake (DPoS) and Byzantine Fault Tolerance (BFT) are emerging to address specific challenges related to computational resources and untrustworthy participants.

Application frameworks like Ethereum, Bitcoin Core, Hyperledger Fabric, Corda, IOTA-Tangle, Quorum, Ripple, and EOSIO have been developed to help build DLT-based applications, each offering unique functionalities tailored to different needs and use cases. These frameworks underpin various industries and sectors, enabling improved efficiency, transparency, and traceability of transactions.

However, the adoption and integration of DLTs are not without challenges. These include scalability issues, particularly related to PoW consensus mechanisms, and the need for ensuring high-security standards dependent on the choice of implementation model (private, consortial, or public). To address these challenges and continue the promotion of DLTs, a collective commitment from industrial actors, institutions, and regulatory bodies is crucial. Furthermore, the development of common standards and norms is essential for the harmonious evolution of these technologies.

The future of DLTs is envisioned to evolve towards hybrid and interoperable solutions, synergizing with other technologies such as Artificial Intelligence and the Internet of Things. This creates potential for digital transformation and innovation across various sectors. As we step into the "post-digital" era, DLTs are expected to underpin our digital interactions, providing an unprecedented level of direct participation and control.

Through this exhaustive analysis of the state of Distributed Ledger Technologies, we hope to provide a comprehensive and informed overview of these revolutionary technologies and stimulate further discussions and research in the field.

KEY FINDINGS

Decentralization at the Forefront: DLTs, like Blockchain and Tangle, are redefining our digital ecosystem by eliminating the need for intermediaries. Their decentralized nature guarantees transparency, security, and immutable record-keeping, opening up a world where control and decisions are distributed rather than centralized.

Innovation meets Challenge: As revolutionary as DLTs are, they aren't without challenges. Scalability, especially in models using Proof of Work consensus, is a hurdle yet to be completely overcome. Moreover, the security of a DLT system is deeply intertwined with its implementation model, emphasizing the importance of informed choices in their deployment.

Interdisciplinary Collaboration: The true potential of DLTs can be harnessed when industries, from finance to healthcare, collaborate. By merging expertise and resources, we can witness the widespread application and integration of DLTs across various sectors, ushering in a new era of digital transformation.

Beyond DLTs – A Synergized Future: DLTs alone are transformative, but when combined with other cutting-edge technologies like Artificial Intelligence and the Internet of Things, the possibilities are boundless. This synergy can lead to unprecedented advancements and solutions for contemporary challenges.

DLTs in the 'Post-Digital' Era: As we transition into an age where digital advancements redefine societal norms and economic paradigms, DLTs are poised to be the backbone of our digital interactions. They promise a future where individuals have direct control and participation, cementing digital trust in an increasingly interconnected world.

WHAT ARE DISTRIBUTED LEDGER TECHNOLOGIES

DLTs, an acronym for Distributed Ledger Technologies, refer to a collection of technologies that facilitate the secure, transparent, and decentralized recording and sharing of information among multiple participants in a network.

DLTs represent the technological infrastructure and types that allow for simultaneous access, validation, and updating of records in a network-linked database. They serve as a foundation for the creation of Blockchain, offering transparency, verifiability, reliability, and controlled access.[1]

The advent of Bitcoin in 2008 marked a significant milestone for DLTs. Blockchain technology quickly rose to prominence due to its resistance to alteration and secure storage of transaction records.[2]

Cryptocurrencies, with their underlying Blockchain technology, have emerged as a significant technological invention in recent times. While cryptocurrencies and Blockchain often dominate discussions, Distributed Ledger Technologies deserve equal attention for their potential to revolutionize conventional commercial transactions. It is essential to understand DLTs and their operation, as many erroneously regard them as synonymous with Blockchain technology.[3]

Key Features of DLT



Immutable

Data once written, cannot be altered.



Shared

The ledger shared among peers, and central authority doesn't own it.



Distributed

Data stored in a distributed format which nodes having full or partial copies.



Append Only

Data added to the ledger in append-only mode.

Source: 101blockchains

Research suggests that global spending on Blockchain-based solutions could reach approximately

$15.9 billion

by 2023

with spending of around

$4.3 billion

in 2020 alone

Source: bitcoin.org & Satoshi Nakamoto

To understand the concept of DLTs, we can compare it to peer-to-peer technologies like email, internet telephony, or file sharing, which operate on the Internet. However, traditional P2P technologies often encounter difficulties in transferring ownership of goods. Bitcoin emerged as a peer-to-peer electronic cash system in 2008, establishing specific approaches to information and transaction organization. Over time, various approaches for organizing Blockchain-based information and facilitating goods transfers in a P2P format have arisen. Consequently, the term DLTs has emerged as an umbrella term encompassing these novel technologies.

DLTs essentially involve a new and continually evolving approach to the recording and sharing of information across multiple data archives. Each data archive (i.e., the ledgers) contains the same data, subject to maintenance and control through a distributed network of computer servers called nodes. Therefore, we can consider DLTs as a distributed database with certain unique properties.[4]

Blockchain is just one type of DLTs that uses cryptographic and algorithmic approaches to create and verify a continuously expanding data structure, where data is added sequentially to form a chain of blocks that function as a ledger.

Nodes initiate new additions to the chain by creating a new "block" of data that contains records of various transactions. Subsequently, information related to the new block is shared throughout the network in the form of encrypted information. Consequently, the Blockchain ensures that transaction details are not publicly available.

Blockchain startups worldwide raised a total of

$4.15 billion

in equity funding

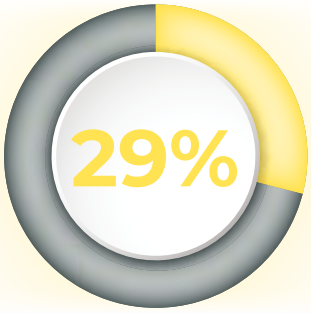
in 2018

Source: bitcoin.org & Satoshi Nakamoto

Subsequently, all network participants evaluate the data block and verify its validity according to a predefined algorithmic validation method, known as the Blockchain consensus mechanism. In the case of Bitcoin, for example, the consensus is valid if it is validated by 50% plus one of the nodes. When the data block is validated, all participants can add the block to their own ledgers, thereby creating a "chain of blocks": the Blockchain.

In summary, DLTs ensure that changes to the ledger are reflected across the network and that all network members have a detailed and identical copy of the entire ledger at any specific moment. It is clear that DLTs capabilities are primarily responsible due to two fundamental components.

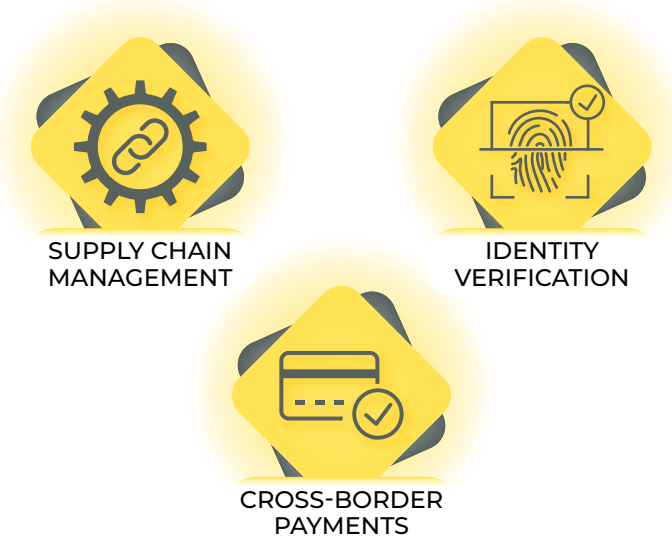
The consumer products and manufacturing industry accounted for the highest share of companies



utilizing Blockchain technology.

Source: bitcoin.org & Satoshi Nakamoto

Blockchain could have a particularly profound impact in areas such as:



Source: BeInCrypto

The first fundamental aspect of DLTs-based systems or infrastructures is the ability to store, record, and exchange digital information among various consenting parties, without the need for any central trust authority or ledger custodian. The second fundamental component of DLTs refers to avoiding double-spending, i.e., sending the same digital asset or token to multiple parties. Without a central control authority, the risk of double-spending is quite high, especially among parties that do not know each other.[5]

DLTs allow for secure and accurate storage of information using cryptography. The data can be accessed using "keys" and cryptographic signatures. Once the information is stored, it can become an immutable database; network rules, integrated into the database programming encoding, govern the ledger.

Because they are decentralized, private, and encrypted, distributed ledgers are less susceptible to cybercrime, as a successful attack would have to simultaneously target all copies stored in the network. Moreover, peer-to-peer sharing and record updating make the entire process much faster, more efficient, and economical.

Each device in a DLTs network keeps a copy of the ledger. These devices are called nodes, and a network can have any number of nodes. Any changes to the ledger, such as the transfer of data from one block to another, are recorded in all nodes. Since each node has a copy of the ledger, each publishes its own version with the most recent transactions.

If the network reaches a consensus on the validity of the most recent ledger, transactions are finalized, encrypted, and used as the basis for subsequent transactions. As we have seen, this is how Blockchain develops: each block contains encrypted information about the previous block, making them impossible to modify.

These data reflect the excitement and growing interest that DLTs are stirring up in various sectors. Their unique characteristics, such as transparency, immunity to manipulation, and the ability to create and manage secure transactions without a central authority, make them extremely attractive to various kinds of organizations.

By understanding the potential of DLTs and exploring their nuances, we can unlock new possibilities for secure and efficient digital asset transfers. Let's embark on this journey to comprehend the power of DLTs.

The global value of Distributed Ledger Technologies (DLTs) is expected to reach



by 2023

Compound annual growth rate of



between 2020 and 2023

Source: Statista

THE CONSENSUS MECHANISMS:
PROOF OF WORK AND PROOF OF STAKE, BUT NOT ONLY

In the previous chapter, we conducted an in-depth examination of what DLTs are and how they function. We discovered that DLTs, such as Blockchain, offer a decentralized approach to recording and sharing information across a network of nodes. However, a crucial aspect to understand about DLTs is the consensus mechanism utilized to ensure the integrity and security of transactions.

In this chapter, we will delve into the two primary consensus modes used in DLTs: Proof of Work (PoW) and Proof of Stake (PoS). These two algorithms are fundamental for transaction validation and maintaining security within a decentralized network.

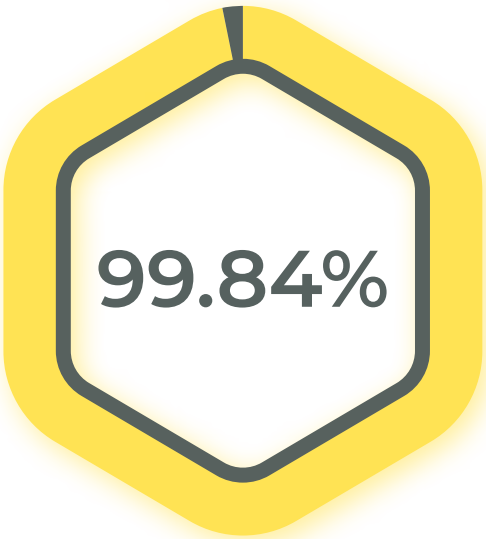
Proof of Work (PoW) and Proof of Stake (PoS) are two consensus mechanisms employed in Blockchain to guarantee the security and validity of transactions. Although both are significant algorithms, they exhibit substantial differences in their implementation and resource requirements.[6]

In Proof of Work, network participants, referred to as "miners," are required to solve complex mathematical problems using a significant amount of computing power. This process demands substantial energy resources, making PoW a resource-intensive system. The first miner to solve the problem is rewarded with cryptocurrency.

Ethereum's switch in 2022, called **'The Merge'** transition from

PoW to PoS

reduced the blockchain's energy consumption by



Source: Digiconomist



The ubiquity of Distributed Ledger Technologies (DLTs) carries enormous potential to reshape our digital world, acting as a powerful catalyst for transformation. Yet, our ability to fully unlock and exploit this potential critically depends on extending its adoption beyond the well-trodden paths of the cryptocurrency sphere. This expansion requires us to conceptualize its application within broader business contexts, particularly by leveraging smart contracts as business process enablers. Such a perspective transforms DLTs from a mere technological marvel into an empowering tool for revolutionizing traditional business operations.

Antonio Grasso
Founder & CEO of Digital Business Innovation

The security of PoW derives from the computational power necessary to manipulate the Blockchain. To alter transactions, a hacker would need to control the majority of the network's computing power, a feat very challenging to achieve.

In contrast, Proof of Stake involves participants pledging a certain amount of existing cryptocurrency, referred to as a "stake," to be selected as validators of new blocks. The probability of being chosen as a validator depends on the number of tokens held. Therefore, the more tokens you own, the greater your chances of being selected for validation. Unlike PoW, PoS requires fewer computational resources, making it less energy-intensive. Stakeholders participating in PoS can also receive rewards in the form of interest on the tokens they stake. The security of PoS is based on the assumption that stakeholders have a vested interest in protecting the network as they own a significant amount of cryptocurrency.

In summary, while PoW requires significant computational power and rewards miners for solving complex mathematical problems, PoS relies on the quantity of tokens held by stakeholders to select block validators. Both algorithms provide a secure system for managing transactions on the Blockchain, but PoS holds advantages in terms of energy efficiency and stakeholder engagement.

While each algorithm has its unique features, some variants have recently emerged that offer additional advantages and improvements.

One such variant of Proof of Stake is Delegated Proof of Stake (DPoS), where the network selects a limited number of validators to validate transactions. This approach significantly reduces the computational resources necessary to secure the network. DPoS is considered a more democratic system for selecting validators and offers enhanced scalability.

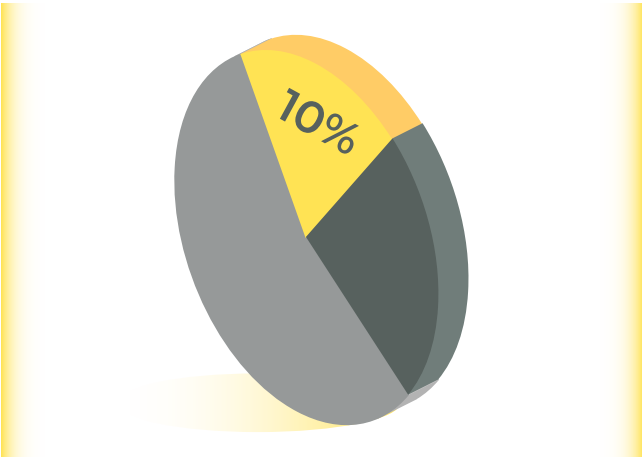
Another consensus mechanism, known as Byzantine Fault Tolerance (BFT), is utilized to achieve consensus within a decentralized network. In this case, validators agree on a consensus value based on a voting system. The primary goal of BFT is to avoid the Byzantine Generals Problem, which pertains to the necessity of reaching a reliable consensus in the presence of potentially faulty or malicious participants.[1]

While Delegated PoS reduces the required computational resources and improves scalability, BFT provides a solution for fault tolerance and security in the presence of untrustworthy participants.

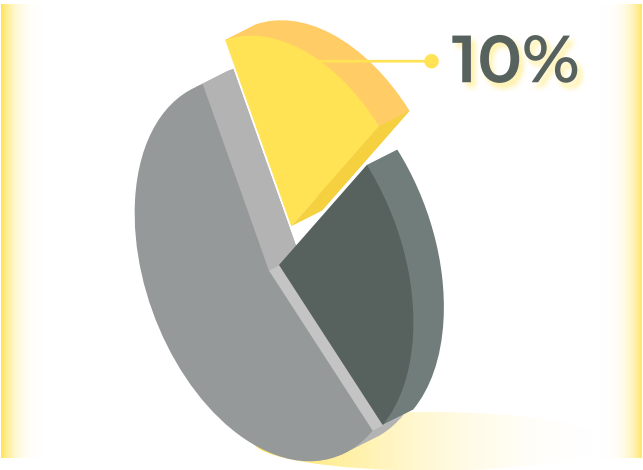
In conclusion, variants such as Delegated Proof of Stake (DPoS) and Byzantine Fault Tolerance (BFT) offer appealing alternatives to Proof of Work (PoW) and Proof of Stake (PoS), introducing improvements in both computational resources and security in the management of transactions within the Blockchain.

A user's reward from their delegate is related to the portion of the total stake they represent. For example:

if a user only makes up



of the entire staking pool, they would receive up to



of the total reward.

Source: crypto.com

THE TYPES OF DLTS

There exist various types of Distributed Ledger Technologies which, as we have seen, represent an innovative and potent infrastructure for managing and recording digital transactions. Among the main DLT solutions, we find the following:

People appear to be optimistic about the future of Blockchain:

86%

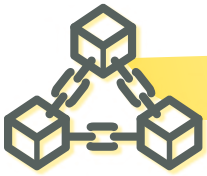
of participants in the survey believe that Blockchain technology will facilitate contactless integration of business processes.

This optimism is well-founded, as approximately:

60%

of Chief Information Officers across all sectors are poised to incorporate Blockchain technology into their infrastructure, as evidenced by statistics on Blockchain growth.

Source: Demand Sage



BLOCKCHAIN

A Blockchain is a decentralized database or ledger shared across multiple nodes within a computer network. Although Blockchain technology is commonly associated with cryptocurrencies, like Bitcoin, its applications go beyond that. Blockchains have the ability to make data immutable, which means they cannot be altered. The immutability of a Blockchain stems from the fact that once a block is added, it cannot be modified. Consequently, the primary trust required in a Blockchain lies at the point of data insertion by users or programs. This eliminates the need for intermediaries such as auditors or humans, who often introduce costs and errors. Since its inception in 2009 with Bitcoin, Blockchain technology has seen significant growth with the emergence of various cryptocurrencies, decentralized finance (DeFi) applications, non-fungible tokens (NFTs), and smart contracts.[7]

A Blockchain consists of scripts or programs that perform the typical tasks of a database: data insertion, access, storage, and archiving.

What distinguishes Blockchain is its distributed nature, meaning multiple copies of the Blockchain are stored on numerous machines, and all copies must match for the Blockchain to be considered valid.

The Blockchain collects transaction information and adds it to a block, which can be viewed as a cell on a spreadsheet containing data.
Once a block has been filled with transactions, the information is encrypted using an algorithm, generating a hexadecimal number known as a hash. The hash is then included in the header of the next block and encrypted together with the block's other information. This creates a chain of interconnected blocks that form the Blockchain.

With a compound annual growth rate of

56.3%

the blockchain industry will be worth

$163.83 billion by 2029

Source: Fortune Business Insight

Fast Transaction:
it can process up to

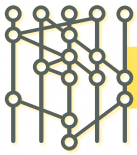


transactions per second

As a result, it is much faster than blockchains like **Bitcoin and Ethereum** and it processes a transaction with



Source: Geeksforgeeks



HASHGRAPH

Hashgraph is a type of Distributed Ledger Technologies that uses a consensus algorithm called 'gossip about gossip' to reach agreement among nodes in a network. Unlike Blockchain, in Hashgraph multiple transactions with the same timestamp can be stored, and a parallel structure is used to store all transactions. Each entry in the ledger is called an 'Event' in this case. No node on the network can manipulate the information or transactions in this Blockchain-less distributed ledger. This means that no one within the DLTs system can change or delay any instructions that are set to occur, nor can they control the transaction process.[8]

Unlike Blockchain, where miners decide which transactions to include in the 'block', the verifier nodes in Hashgraph must include the transactions of both the user and the other user in the same order in which they were made, so that no one is left behind.
What's different about Hashgraph is that all network transactions are provable in this distributed ledger implementation.

UNIQUE SELLING POINTS OF HEDERA HASHGRAPH?

SPEED

Supports thousands of transactions per second, offering faster processing than traditional blockchains.

LOW FEES

The platform offers low transaction fees, making it cost-effective for various use cases.

ECO-FRIENDLY

Consumes significantly less energy compared to blockchain networks like Bitcoin or Ethereum.

Source: House of Chimera

As soon as a transaction takes place on the network, everyone on the network will know where the transaction will be recorded in the ledger within a few minutes. Furthermore, everyone on the network will be aware that the transaction has been detected by the entire network, allowing them to make the necessary adjustments. This indicates that nodes will make the changes before discarding the transaction. This information doesn't need to be stored in the ledger indefinitely. As a result, the Hashgraph database platform only requires a few gigabytes of storage space to store all of its data.

TEMPO



TEMPO is a consensus algorithm developed by Radix DLT. This is a consensus protocol designed to ensure scalability and efficiency in Blockchain or DLT networks. The primary objective of TEMPO is to allow swift finalization of transactions and reliable confirmation of data.[9] Unlike some other consensus algorithms, TEMPO relies on the random selection of validators who participate in the transaction confirmation process. This random selection helps reduce the centralization of decision-making power and enhances resistance to manipulation. One of the key advantages of TEMPO is its ability to handle a high number of transactions per second without compromising security or decentralization. This makes it suitable for applications that require large scalability, such as digital payments or asset exchange networks.

TEMPO FEATURES

SHARDING

Every node on the network keeps a shard (smaller section of the global ledger) with a unique ID.

GOSSIP PROTOCOL

Nodes broadcast all their information to synchronize their own shards.

LOGICAL CLOCKS

Rather than timestamping the transactions, this DLT remembers the sequence of the transactions to reach consensus.

Source: 101blockchains

MINING

The previous transaction validates the succeeding to achieve consensus.

TRANSACTIONS PER SECOND

Unique data structure via directed acyclic graphs ensures that scalability and TPS are high.

DATA STRUCTURE

Data structure follows the directed acyclic graph mechanism where each transaction is independent.

VALIDATION OF TRANSACTIONS

The success of present transaction relies on its ability to validate two previous transactions.

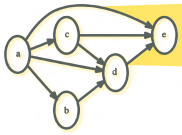
TIME OF LAUNCH

NXT is the first platform utilizing DAG, and it came out on November 9, 2015.

NETWORKS RUNNING ON THE PLATFORM

NXT, Tangle, and ByteBall are the most popular networks using DAG foundation.

Source: 101blockchains



DIRECTED ACYCLIC GRAPH (DAG)

Directed Acyclic Graph (DAG) is a sophisticated addition to distributed ledger technologies that eschews the traditional Blockchain structure. It was developed as an alternative to Blockchain-based Distributed Ledger Technology (DLT).[8] Without relying on a Blockchain, DAG offers the benefits of Blockchain and advances them further.

While it may serve as an alternative, DAG's structure is notably distinct. A significant advantage of DAG's ledger system is its capability to handle fee-less Nano transactions, a feature made possible by the improved scalability of the network as it grows.

In DAG's consensus mechanism, transactions reside on nodes, much like they do in Blockchain systems. Each participant in this network is termed a 'node'. Every node in the network validates the ledger, and once validated, these transactions are marked as such. While any node can initiate a transaction, it must first authenticate at least two prior ledger transactions to validate its own. The confirmation of a node's transaction is dependent on its previous validation activities. Thus, a transaction backed by a lengthy chain of previously authenticated transactions gains the most prominence in the ledger.

Furthermore, to prevent members from only validating their own transactions, an algorithm will randomly select two prior transactions for each node's validation. This method introduces a novel way of achieving enhanced scalability via consensus. For enterprises needing high transaction throughput, this ledger system is particularly well-suited due to its inherent characteristics.

HOLOCHAIN

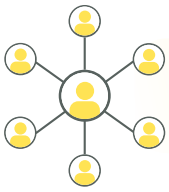


Holochain is an open-source framework designed for the development of serverless applications and functions as a peer-to-peer network protocol. It empowers users to run applications on individual devices, ensuring user data sovereignty and direct communication between participants. Unlike the traditional Blockchain that relies on a unified data structure of linked blocks, Holochain employs a distinct method termed a 'hash chain' for data recording and sharing.

Each participant on Holochain maintains a unique ledger, referred to as a Distributed Hash Table (DHT), where data is stored locally.[10] Digital signatures validate this data, and consistency across multiple ledgers is achieved through selective information dissemination. This decentralization enhances scalability, diminishes the need for global consensus, and grants users the ability to control their data autonomously.

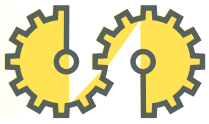
In essence, while Blockchain is characterized by its specific data structure and consensus method, Holochain presents an innovative perspective on decentralized data recording and sharing, emphasizing the creation of independent peer-to-peer applications that can also interface with other networks or Holochain instances.

Holochain Use Cases



Social networks

Supply chains





P2P networks

Collaborative applications



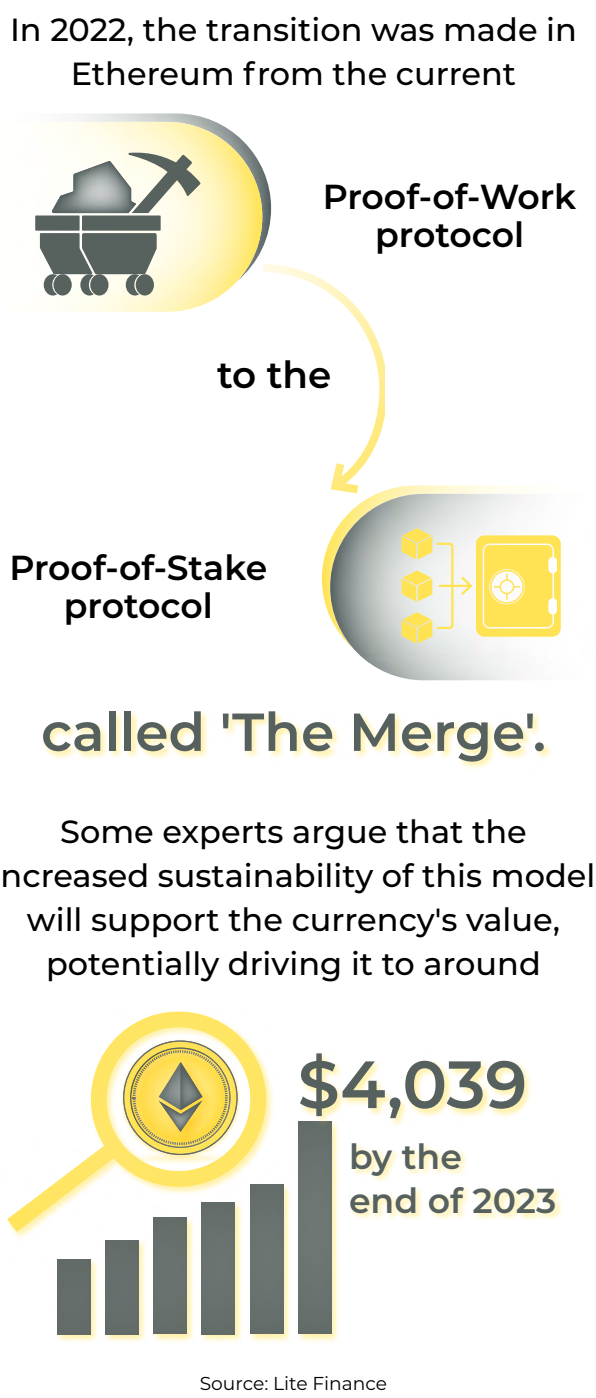


Rating platforms

Source: 101blockchains.com

APPLICATION FRAMEWORKS

There are several application frameworks that can be used to develop applications based on Distributed Ledger Technologies. Some of the most popular frameworks include:



Source: Lite Finance

ETHEREUM

Ethereum is a leading decentralized blockchain platform, known for executing and verifying sophisticated Smart Contracts or dedicated application codes. It leverages a strong peer-to-peer framework, allowing users to conduct transactions without centralized control[11]. Once a transaction is completed, it's indelibly recorded, verified, and securely distributed across the network, guaranteeing both autonomy and traceability for users. To participate, individuals must set up Ethereum accounts. Transactions require the sender's authentication through a signature and the spending of Ether, Ethereum's inherent cryptocurrency, as a transaction fee. The integration of Ethereum's unique scripting language, Solidity, alongside the Ethereum Virtual Machine, positions Ethereum as a highly adaptable platform for creating decentralized applications. Developers looking to utilize Ethereum's smart contract capabilities are introduced to a polished protocol environment replete with a thorough array of developmental resources and proven best practices[12]. A significant milestone for Ethereum was "The Merge", which marked the union of Ethereum's primary execution layer (Mainnet) with its innovative Proof of Stake layer, the Beacon Chain. This merger eliminated energy-intensive mining processes, shifting the focus to using ETH for staking, highlighting Ethereum's dedication to enhanced scalability, superior security, and environmentally-friendly operations[13].

There are

120.23 million

Ethereum in circulation as of June 2023.

Source: Coinmarketcap

Furthermore, the Ethereum environment offers an instinctive user experience thanks to built-in wallets like MetaMask, Argent, and Rainbow. These tools simplify interactions with the Ethereum Blockchain and its array of smart contracts. Ethereum's broad user base attracts developers, consolidating its position as the chief hub for decentralized applications. Looking ahead, the forthcoming launch of Ethereum 2.0, a backward-compatible system in its latter development stages, ushers in a promising phase. It anticipates increased scalability, paving the way for the birth of decentralized applications with faster transaction speeds.

BITCOIN CORE

Bitcoin Core is the open-source software that serves as the official client for the Bitcoin network. It is the reference software for the Bitcoin protocol and represents the primary implementation of the software used by most users and operators within the Bitcoin network. Bitcoin Core offers a comprehensive range of features to participate in the Bitcoin network.[14]

It includes a digital wallet that allows users to send, receive, and manage their Bitcoin transactions. The software synchronizes the entire Bitcoin Blockchain, which is a public ledger of all transactions made on the network, enabling users to independently verify transactions and ownership of their Bitcoins.

In addition, Bitcoin Core also supports the mining function, which is the process through which new bitcoins are created, and transactions are confirmed and inserted into the Blockchain.

Users can contribute to the security and reliability of the Bitcoin network by offering their computing power for mining. Bitcoin Core is developed by a team of volunteer developers who work collaboratively with the Bitcoin community. The software undergoes continuous revisions, updates, and enhancements to ensure its security, stability, and optimal functionality.

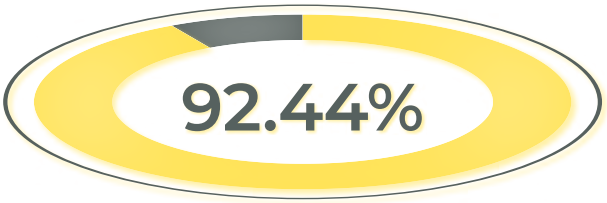
Currently, there are just over

19.4 million

Bitcoins in circulation.



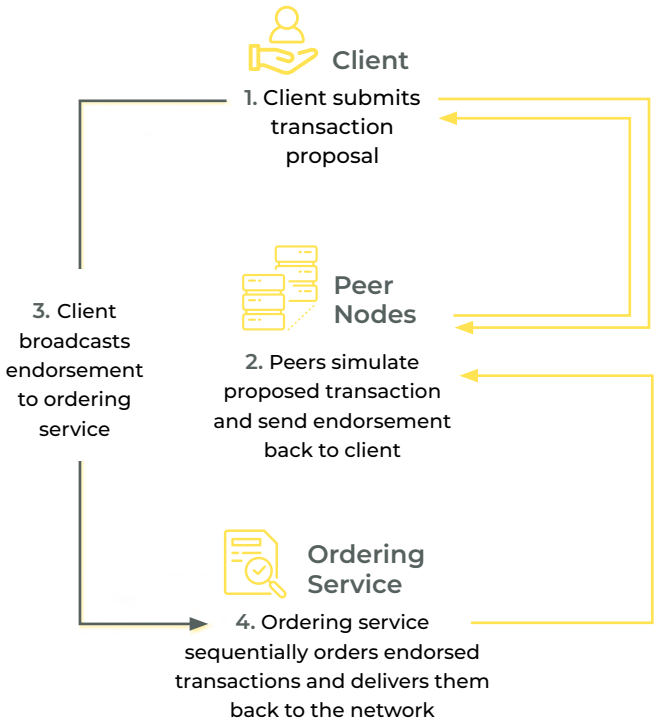
This means that out of the total supply of 21 million Bitcoin tokens



are already in circulation.

Source: CoinMarketCap

How does Hyperledger Fabric work?



Source: AWS

HYPERLEDGER FABRIC

Hyperledger Fabric, initiated by the Linux Foundation in 2015, is an open-source Blockchain framework. Designed with a focus on modularity and versatility, it offers distinctive features for identity management and access control. These characteristics make it suitable for various industrial applications, including supply chain traceability, trade finance, loyalty programs, as well as settlement and clearing of financial activities.[15]
A Hyperledger Fabric network is made up of unique organizations or members that interact within the network, such as financial institutions in a banking network or shipping partners in a supply chain network.

CORDA

Corda is a distributed ledger software designed to facilitate data processing and recording in a decentralized network environment. The structure of smart contracts within the Corda R3 architecture involves an agreement that is based on computer code but incorporates human oversight and input. Furthermore, these smart contracts can be legally applied when necessary or desired. In cases where an injustice is perceived within the smart contract function of Corda's private Blockchain, individuals have the possibility to resort to legal action and undertake appropriate measures.

KEY TAKEAWAY

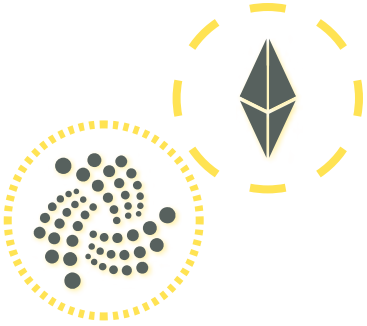
Smart Contracts are self-executable digital protocols that facilitate, verify, and automatically enforce agreements between parties without the need for intermediaries. They are immutable, transparent, and secure thanks to Blockchain technology. Smart Contracts allow for the automation of transactions, the enactment of complex business logics, and the elimination of the risk of tampering, thereby enhancing efficiency and trust in business operations.

The Bitcoin Blockchain can handle roughly

5 transactions per second (TPS)

although this may vary at times.

For Ethereum,
it's typically around
15 TPS



For IOTA,
it's possible for its network to handle up to around
1,000 TPS

Source: Forbes



IOTA-TANGLE

IOTA-Tangle is a Distributed Ledger Technology (DLT) platform focusing on the Internet of Things (IoT). Unlike other cryptocurrencies like Bitcoin, IOTA distinguishes itself by employing a distributed ledger technology that veers away from the Blockchain. IOTA relies on a proprietary technology called Tangle, a consensus algorithm that necessitates users to validate two transactions to finalize their own IOTA transactions. From a technical perspective, Tangle is a consensus algorithm based on a Directed Acyclic Graph (DAG).[16]

This approach neither involves miners nor validators and does not utilize blocks or transaction fees.
As stated on IOTA's website, this allows the cryptocurrency to "overcome the challenges related to the cost and scalability of Blockchain".[17]
The DAG structure was designed to solve scalability issues and the costs associated with Blockchain, thereby allowing for feeless transactions, lower computational costs, and the possibility to connect with IoT devices. IOTA's structure is particularly appealing for the IoT economy as it removes barriers constituted by high transaction costs, which would necessitate monitoring and human intervention, contrary to what the IoT strives to accomplish.[18]
IOTA's Smart Contracts platform is currently revising its ledger technology and consensus protocol. IOTA version 2.0 will be fully decentralized.

QUORUM

Quorum is a Distributed Ledger Technology (DLT) framework developed by J.P. Morgan, based on Ethereum. Its primary aim is to provide an authorized implementation of Ethereum's Blockchain that supports the privacy of contracts and transactions.[19]

The distinctiveness of this network lies in its new function called the "private transaction identifier," which guarantees data confidentiality. The primary objective of Quorum's development is the extensive use of existing technologies. Therefore, even in the case of future updates to the Ethereum network, Quorum's Blockchain will remain unchanged to maintain synchronization between the two.

QUORUM ARCHITECTURE

Three key components

QUORUM NODE

A command line tool based on Geth.

CONSTELLATION TRANSACTION MANAGER

It takes care of the transaction data until it gets completed.

ENCLAVE

Enclave handles the sensitive information where the Transaction Manager delegates key functions such as encryption/decryption.

Source: 101blockchains

EOSIO Features

PERFORMANT

- Subsecond block time
- State stored in multi index tables
- EOSVM WASM engine (12x speed increase over EOSIO 1.0)

FLEXIBLE

- Update Smart Contracts or add binary extensions
- Network behaviour defined by system contracts which are highly configurable

ACCESSIBLE

- Smart Contracts written in C++
- Wide range of integration SDKs
- EOSIO development tools (more later...)

Source: Learn about Blockchain & Eosio, eos.io

EOSIO

EOSIO is a distinct DLT (Distributed Ledger Technology) platform, fundamentally relying on a delegated consensus model. This model is instrumental in facilitating scalable transactions while maintaining high performance. At its core, EOSIO extends a development framework, aiding the creation of decentralized applications (dApps) which can be crafted using languages including C++, Rust, and JavaScript. Furthermore, it demonstrates a high degree of configurability, catering to both public and private blockchain environments with the potential for custom governance through smart contracts. In addition to its technological strengths, EOSIO adopts advanced web security standards, emphasizing end-to-end authentication, thereby underscoring its commitment to data integrity. Its design flexibility also means it's aptly suited to meet various compliance and regulatory needs. [20]

XRP-RIPPLE

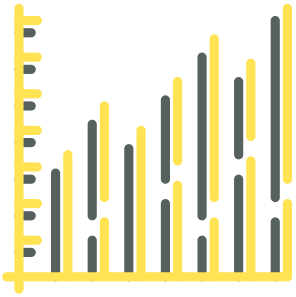
Ripple, a Blockchain-based digital payments network and protocol, also has its own cryptocurrency called XRP. Ripple's main process is a payment settlement and money transfer asset exchange system, akin to the SWIFT system used by banks and financial intermediaries for international money and securities transactions. The token used for the cryptocurrency is pre-mined and uses the trading symbol XRP. Ripple is the name of the company and network, while XRP is the cryptocurrency token. The purpose of XRP is to serve as an intermediary exchange mechanism between two currencies or networks, acting as a sort of temporary settlement layer. Ripple was first released in 2012 and was co-founded by Chris Larsen and Jed McCaleb. Notably, Ripple uses a consensus mechanism through a group of servers owned by banks to confirm transactions, rather than relying on traditional Blockchain mining. Ripple transactions consume less energy than Bitcoin transactions, are confirmed in a few seconds, and have significantly lower costs, while Bitcoin transactions require more energy, take longer to confirm, and involve higher transaction fees. Ripple (XRP) ranks among the most valued cryptocurrencies based on market capitalization. Although the Ripple payment system is primarily intended for banks, individual investors have the opportunity to speculate on the price of XRP.[21]

As of 29 May 2023,
one XRP was worth around

$48.13 cents



Ripple was the sixth-largest cryptocurrency with a market cap of more than



$25 billion

Source: CoinMarketCap

CONCLUSIONS

This research aims to provide a comprehensive and up-to-date analysis of the state of Distributed Ledger Technologies (DLTs) and their impact on the current technological landscape. Through a critical review of the various implementations and applications of DLTs, a clear trend towards increased adoption and integration of these technologies is emerging.

DLTs, with their decentralized and immutable nature, offer innovative solutions for data management and sharing, while simultaneously ensuring security, transparency, and reliability. Technologies such as Blockchain, Tangle, and Hashgraph have proven their utility in creating reliable digital ecosystems, eliminating traditional intermediation, and enabling the creation of new business and governance models.

However, despite the great potential of DLTs, several challenges still persist. Scalability remains a critical issue in some areas related to the Proof of Work consensus, with the need to develop efficient solutions to handle the increasing volume of transactions and participants.

Moreover, it is important to consider that the security of DLTs is closely tied to the chosen implementation model, which can be private, consortial, or public. The choice of approach adopted in the implementation stage plays a fundamental role and cannot be overlooked. A careful assessment of the mode of technology usage is of critical importance to ensure an adequate level of security.

To continue promoting the development and adoption of DLTs, a collective commitment from industrial actors, institutions, and regulatory bodies is necessary. This commitment must be guided by the fundamental principle of decentralization, which is one of the pillars driving DLTs adoption. The aim is to create an ecosystem where decisions and control of transactions are not concentrated in a single central entity, but are distributed among the network participants.

To achieve this aim, significant investments in research and development must be dedicated, with the purpose of improving existing DLT solutions and promoting innovation. Additionally, collaborations between various industrial sectors, such as finance, energy, and healthcare, can foster the spread and application of DLTs across different domains.

At the same time, the development of common standards and norms is crucial to ensuring the evolution and stability of these technologies. The harmonization of DLTs usage practices and protocols promotes interoperability between different platforms and enhances the trust of users and involved actors.

Five Transformative Uses of Distributed Ledger Technology (DLT)



Source: Investopedia

Therefore, the future of such technologies appears promising, evolving towards hybrid and interoperable solutions that combine the best features of different technologies.

The synergy of DLTs with other technologies, such as Artificial Intelligence and the Internet of Things, opens up new possibilities for digital transformation and innovation in a variety of sectors.

As we approach the "post-digital" era, which could rightly be defined as a period of social and economic revolutions surpassing our most optimistic predictions, it is important to recognize the fundamental role of Distributed Ledger Technologies (DLTs). In this transformative phase, DLTs will become the backbone of our digital interactions, providing individuals with an unprecedented level of direct participation and control. We can no longer remain mere observers of this metamorphosis; instead, we must take center stage and become the architects of this transformation.

What DLTs offer us is a futuristic vision where the individual is at the center, and digital trust is ensured. This is a theme that requires further exploration to understand the implications and opportunities offered by these technologies and how we can adapt to this new reality. In light of these intriguing discoveries, the horizon of our studies further expands, paving the way for a fascinating future of exploration and deepening. We invite you to stay connected with us, ready to jointly explore new frontiers of knowledge.

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